

STRUCTURE OF THE ATOM

Class 9 | Science | Chapter 4

SUB-ATOMIC PARTICLES

Atoms are made up of three basic particles: electrons, protons and neutrons.

Particle	Charge	Location	Relative Mass
Electron (e-)	Negative	Outside nucleus, in shells	~1/1836 (negligible)
Proton (p+)	Positive	Inside nucleus	1 u
Neutron (n)	Neutral	Inside nucleus	1 u

Discovered by: Electron -- J.J. Thomson; Proton -- E. Goldstein; Neutron -- J. Chadwick.

EARLY ATOMIC MODELS

Thomson's Model

'Plum pudding' model -- positive sphere with electrons embedded like plums in a pudding.

Rutherford's Model

Gold foil experiment -> most of atom is empty space; nucleus is small, dense & positively charged; electrons revolve around it.

Bohr's Model

Electrons revolve in fixed, discrete circular orbits/shells (K, L, M, N...) each with a fixed energy -- energy is not radiated while in that orbit.

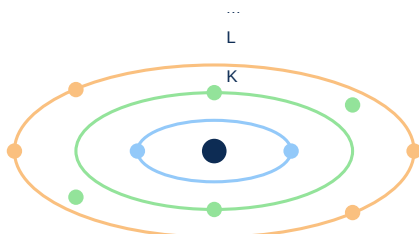
DISTRIBUTION OF ELECTRONS

Bohr-Bury Scheme & Valency

BOHR-BURY SCHEME (SHELL FILLING RULES)

- Maximum electrons in a shell = $2n^2$ (n = shell number: K=1, L=2, M=3, N=4).
- So K-shell max 2, L-shell max 8, M-shell max 18, N-shell max 32.
- Outermost shell cannot hold more than 8 electrons (octet rule).
- A shell is not filled until the inner shells are filled first.

DIAGRAM: SHELLS AROUND NUCLEUS (K, L, M)



VALENCE ELECTRONS & VALENCY

Electrons in the outermost shell are called valence electrons -- they decide the valency of the element.

Ex: Sodium (2,8,1) -> 1 valence electron -> valency 1. Oxygen (2,6) -> valency 2 (8-6).

EXAMPLE ELECTRON DISTRIBUTIONS

Element	Atomic No.	Distribution
Hydrogen	1	1
Carbon	6	2,4
Sodium	11	2,8,1
Chlorine	17	2,8,7
Argon	18	2,8,8

ATOMIC NUMBER, MASS NUMBER & ISOTOPES

Key Definitions & Examples

ATOMIC NUMBER (Z)

Number of protons present in the nucleus of an atom (= number of electrons in a neutral atom).

MASS NUMBER (A)

Sum of the number of protons and neutrons in the nucleus of an atom.

Mass Number (A) = Protons (p) + Neutrons (n)

ISOTOPES

Atoms of the same element having the same atomic number but different mass numbers (different number of neutrons).

Ex: Hydrogen isotopes -- Protium ($1\text{H}1$), Deuterium ($2\text{H}1$), Tritium ($3\text{H}1$). Carbon-12 & Carbon-14.

Uses of isotopes: U-235 in nuclear reactors, I-131 in treatment of goitre, Co-60 in cancer treatment.

ISOBARS

Atoms of different elements having the same mass number but different atomic numbers.

Ex: Calcium (atomic no. 20) and Argon (atomic no. 18), both with mass number 40.

SUMMARY & REVISION

Key Terms + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Electron** -> negatively charged, revolves around nucleus
- **Proton** -> positively charged, inside nucleus
- **Neutron** -> neutral, inside nucleus
- **Atomic number** -> number of protons
- **Mass number** -> protons + neutrons
- **Isotopes** -> same atomic no., different mass no.
- **Isobars** -> same mass no., different atomic no.

SOLVED EXAMPLE

Q: An atom has 17 protons and 18 neutrons. Find its atomic number and mass number.

A: Atomic number = protons = 17. Mass number = 17 + 18 = 35.

ONE-LINE RECAP FOR REVISION

Atom = nucleus (protons + neutrons) + electrons in shells.

Max electrons per shell = $2n^2$; outermost shell max 8.

Isotopes: same protons, different neutrons. Isobars: same mass no., different protons.

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